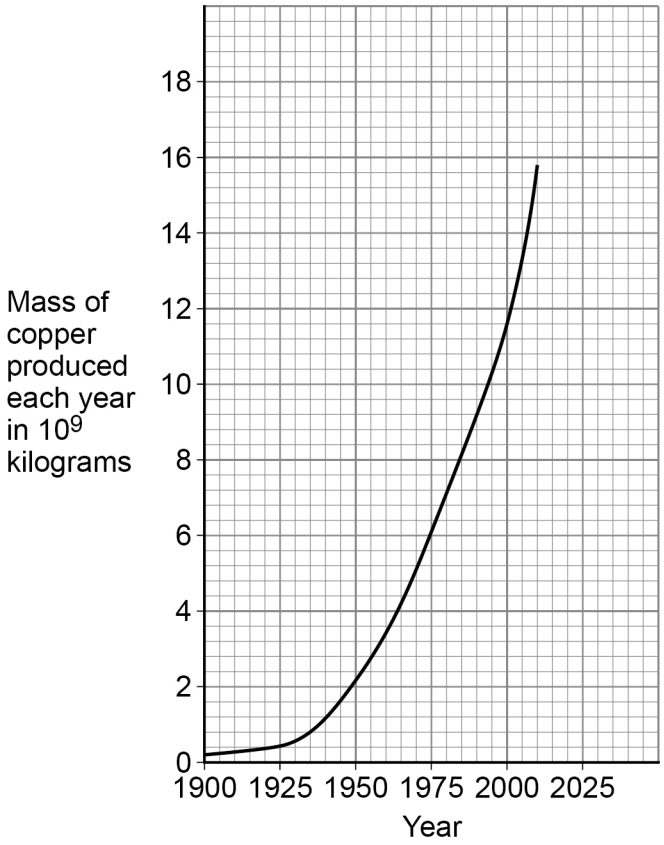
Do not write outside the box

0 1 Copper is a useful metal.

0 1 . 1 **Figure 1** shows the mass of copper produced between 1900 and 2010.

Figure 1



Give **two** conclusions that can be made from **Figure 1**.

[2 marks]

1 _____

_____

2 _____

_____

Question 1 continues on the next page

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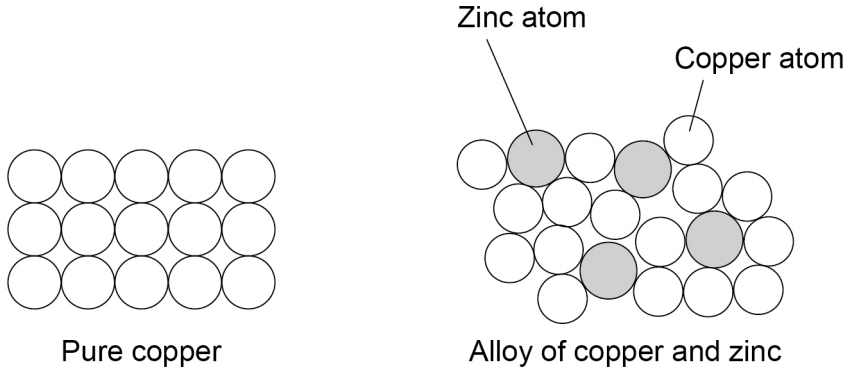
Mixtures of copper and zinc are heated to produce alloys.

0 1 . 2 **Figure 2** represents the structures of pure copper and of an alloy of copper and zinc.

Figure 2



Explain why the alloy of copper and zinc is harder than pure copper.

Use **Figure 2**.

[3 marks]

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0 1 . 3 A 5.25 g sample of an alloy of copper and zinc contains 13.5% zinc by mass.

Calculate the mass of **copper** in the 5.25 g sample.

Give your answer to 3 significant figures.

[4 marks]

Mass of **copper** (3 significant figures) = _____ g

9

Turn over for the next question